

Author	Description	Methodology	Results	Limitations
FFT findings				
Nicolas Vasilache, Jeff Johnson, Michael Mathieu, Soumith Chintala, Serkan Piantino, Yann LeCun	FFT-based convolution implementations for CNNs using cuFFT and custom fbfft library, focusing on GPU performance optimization.	Implemented FFT convolution using cuFFT library with autotuning strategy, developed custom fbfft implementation with warp-level optimization for batched 1-D and 2-D FFTs.	cuFFT implementation: 1.4x-14.5x speedup over cuDNN for common layer sizes, up to 23.54x for 13x13 kernels. fbfft: 1.5x speedup over cuFFT, achieves 78% efficiency at 97.5% occupancy.	Poor performance for small kernels (3x3) due to overhead. fbfft limited to square convolutions with power-of-2 sizes. Limited size range (8-64) for maximum gains.
Michael Mathieu, Mikael Henaff, Yann LeCun	Fast Training of Convolutional Networks through FFTs - An algorithm that accelerates CNN training and inference by computing convolutions as pointwise products in the Fourier domain while reusing transformed feature maps multiple times.	Transform input feature maps and kernels to frequency domain using FFT. Perform all convolutions as element-wise multiplications in Fourier domain. Apply inverse FFT to get spatial domain outputs. Reuse FFT computations across multiple convolutions in forward pass, backward pass, and gradient computation. Developed custom CUDA FFT implementation optimized for many small transforms.	Significant speedups over CudaConv and Torch7 implementations across all operations. Best performance on accGradParameters operation (most computationally expensive). Complete network training iteration: 867ms vs 1756ms (Torch7) and 2495ms (CudaConv) - approximately 2-3x speedup. Performance independent of kernel size, enabling use of larger kernels. Accuracy differences within acceptable range (10^{-5} to 10^{-4}).	Requires additional GPU memory to store frequency domain representations. Input images not power-of-2 must be padded (e.g., 34x34 padded to 64x64), causing suboptimal performance. Method most effective for large numbers of feature maps - overhead not worthwhile for small networks. Memory requirements increase with larger convolution layers. Current implementation limited to power-of-2 sizes for optimal performance.
Noora Sassali	Deep Learning with Fourier Transformed Images - Evaluates how CNN backbone learns image classification when trained with Fourier-transformed images instead of spatial RGB images, comparing DFT pre-processed images against traditional RGB image training.	Created two EfficientNetV2-S models: one trained with spatial RGB images, another with DFT pre-processed images. Used reduced ImageNet 2012 dataset (64x64 resolution, 1.2M images, 1000 classes). Applied 2D FFT to RGB images creating 6-channel input (real/imaginary parts for each RGB channel). Trained both models from scratch for 10 epochs using SGD optimizer with Cross-Entropy loss.	RGB model achieved 20.37% top-1 accuracy while DFT model reached only 9.19% top-1 accuracy after 10 epochs. Baseline pre-trained model achieved 56.0% top-1 accuracy. RGB model consistently outperformed DFT model across all training phases. Both models failed to classify 61 classes at top-1 accuracy.	Limited to only 10 training epochs due to computational constraints. Used reduced 64x64 resolution instead of full ImageNet resolution. Small batch size (64 vs recommended 128) due to memory constraints. EfficientNetV2-S not optimized for 64x64 input resolution. Batch normalization layers may not be suitable for DFT data.

Frequency domain findings				
Sophie Totterstrom	It investigates whether transforming RGB images to frequency domain using Fourier transforms can improve CNN classification performance compared to spatial domain approaches.	Implemented AlexNet architecture for both RGB and frequency domain images using CIFAR-10/100 datasets. Applied DFT to convert RGB images into 6-channel inputs (3 magnitude + 3 phase spectra). Trained models for 25 epochs and compared classification accuracy between spatial and frequency domain approaches.	RGB models significantly outperformed frequency domain models: CIFAR-10 (RGB: 79.23% vs FFT: 38.78%), CIFAR-100 (RGB: 47.80% vs FFT: 15.84%). Frequency domain models achieved only half the accuracy and showed severe overfitting issues after initial training epochs.	CNNs lose benefits when applied to frequency domain since geometric information is already extracted by Fourier transform. Convolution operations on frequency domain only increase computational complexity without additional value. Architecture not optimized for frequency domain - shallow networks might be more suitable. Limited to small datasets due to CPU-only training constraints.
Wenlin Chen, James Wilson, Stephen Tyree, Kilian Q. Weinberger, Yixin Chen	Frequency-Sensitive Hashed Nets (FreshNets) - A novel CNN compression method that exploits redundancy in convolutional and fully-connected layers by converting filter weights to frequency domain using DCT and applying frequency-sensitive hashing for parameter sharing.	Convert CNN filter weights to frequency domain using Discrete Cosine Transform (DCT). Apply hash-based parameter sharing where frequency components are randomly grouped into hash buckets sharing single values. Allocate fewer hash buckets to high-frequency components (less important) and more to low-frequency components. Learn shared parameters via standard backpropagation with gradient computation in frequency domain.	Evaluated on 8 datasets (CIFAR10/100, SVHN, MNIST variants) with compression ratios 1/16 and 1/64. FreshNets consistently outperformed baselines: CIFAR10 at 1/64 compression achieved 30.79% error vs 43.08% (HashedNets) and 34.35% (LRD). Maintained filter smoothness better than spatial domain methods. Frequency-sensitive scheme (more low-frequency preservation) outperformed uniform compression.	Method adds computational overhead from DCT/inverse DCT operations during training and inference. Performance still degrades significantly at high compression ratios. Requires careful tuning of frequency-sensitive compression rates for optimal performance. Limited evaluation to relatively small-scale datasets and network architectures.
Kai Xu, Minghai Qin, Fei Sun, Yuhao Wang, Yen-Kuang Chen, Fengbo Ren	Learning in frequency domain using DCT coefficients as input to CNNs, with learning-based frequency channel selection to identify important frequency components for image classification.	Convert RGB to YCbCr then DCT domain, group same-frequency components into channels (192 total), use learning-based gate module with Gumbel Softmax for dynamic channel selection.	ImageNet classification: +1.60% accuracy on ResNet-50, +0.63% on MobileNetV2. COCO segmentation: +0.8% AP on Mask R-CNN. Can prune up to 87.5% frequency channels with minimal accuracy loss.	Counter-intuitive observation that fewer channels sometimes perform better suggests high-frequency noise. Limited analysis of frequency-spatial mapping for explainability.

Ee Fey Goh, ZhiYuan Chen, Wei Xiang Lim	Frequency Domain Convolutional Neural Network (FDCNN) for large image classification using FFT-based convolution operations.	Proposed Frequency Domain Convolution (FDC) and Frequency Domain Pooling (FDP) layers using RFFT, kernel initialization in spatial domain then transformed to frequency domain, Channel Independent Convolution (CIC), and convolution artifact removal. Built FDCNN architecture and modified VGG16 with Full FDC layers.	FDCNN achieved 54.21% faster training and 70.74% more memory efficient than equivalent CNN. Full FDC layer 46% faster than conventional convolution at 256/512 channels.	Lack of suitable frequency domain activation function limits architecture design. FDCNN shows accuracy decrease when overfitting begins.
Grad-Cam findings				
Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra	Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization - A technique for producing visual explanations from CNN-based models by using gradients flowing into final convolutional layer to create coarse localization maps highlighting important regions for predictions.	Uses gradients of target concept (class scores) flowing into last convolutional layer to compute neuron importance weights via global average pooling. Performs weighted combination of forward activation maps followed by ReLU to obtain class-discriminative localization maps. Combines with Guided Backpropagation to create high-resolution Guided Grad-CAM visualizations.	Outperformed baselines on ILSVRC-15 localization (56.51% vs 61.12% top-1 error). Human studies showed 61.23% accuracy in class discrimination vs 44.44% for Guided Backpropagation. Achieved 0.254 rank correlation with occlusion maps (higher faithfulness than existing methods). Successfully applied to image classification, captioning, and VQA tasks. Helped identify dataset biases and model failure modes.	Produces coarse localization maps at convolutional feature map resolution (14×14 for VGG/AlexNet). Requires combination with other methods (Guided Backpropagation) for high-resolution visualizations. ReLU operation may miss some relevant negative contributions. Limited to explaining decisions rather than overall model behavior.
Aditya Chattopadhyay, Anirban Sarkar, Prantik Howlader, Vineeth N Balasubramanian	Grad-CAM++ - Improved Visual Explanations for Deep Convolutional Networks. A generalized method that provides better visual explanations of CNN predictions through improved object localization and handling multiple object instances, building upon Grad-CAM with pixel-wise weighting of gradients.	Uses weighted combination of positive partial derivatives of last convolutional layer feature maps with respect to class scores. Derives closed-form solutions for pixel-wise weights using second and third-order derivatives. Applies exponential/softmax activation functions for smooth differentiation. Combines with Guided Backpropagation for high-resolution Guided Grad-CAM++ visualizations.	Outperformed Grad-CAM on objective metrics: 36.84% vs 46.56% average drop, 70.72% vs 29.28% win percentage on ImageNet. Human studies showed 109.69 vs 56.08 preference score for Grad-CAM++. Better localization performance: 0.38 vs 0.28 IoU on PASCAL VOC. Successfully applied to image captioning, 3D action recognition, and knowledge distillation with 35.5% mAP improvement. Works across VGG-16, AlexNet, ResNet-50 architectures.	Requires computation of higher-order derivatives increasing complexity. Performance improvement varies across different architectures and datasets. Still produces coarse localization maps requiring upsampling. Limited to positive gradients assumption which may miss some relevant features. Computational overhead compared to Grad-CAM though still single backward pass.

Hicham Moujahid, Bouchaib Cherradi, Mohammed Al-Sarem, Lhoussain Bahatti, Abou Bakr Assedik, Mohammed Yahya Eljialy, Abdullah Alsaeedi, Faisal Saeed	Combining CNN and Grad-CAM for COVID-19 Disease Prediction - A visual explanation framework using pre-trained CNNs and Grad-CAM for automatic classification and interpretation of COVID-19, pneumonia, and normal chest X-ray images.	Collected and merged real, publicly available X-ray images into a new balanced dataset for three classes. Used pre-trained VGG16, VGG19, and MobileNetV2 models with transfer learning, adding dense and dropout layers. Grad-CAM applied to highlight key image regions influencing model predictions, aiding interpretation and model debugging.	Tuned VGG19 achieved best performance: 96.97% accuracy, 100% precision and F1-score, 99% recall. VGG19 and VGG16 visual explanations corresponded to medically relevant lung areas, confirming reliable predictions. No data augmentation used, all images from validated sources. Outperformed or matched several cited methods on comparable metrics.	Dataset size for COVID-19 cases remains limited and class imbalance possible. Further improvement needs more positive samples and hyperparameter optimization. Some models (e.g., MobileNetV2) showed overfitting and unreliable focus in Grad-CAM visualizations.
Jose Antonio Marmolejo-Saucedo, Utku Kose	Numerical Grad-CAM Based Explainable CNN for Brain Tumor Diagnosis - Proposes numGrad-CAM, an enhanced Grad-CAM variation adding numerical focus area feedback to heatmaps, improving the explainability of CNN decisions for medical image diagnosis.	Developed a 7-layer CNN model for brain tumor diagnosis using the BRATS 2017 MRI dataset (200 images, 170 train, 30 test). Integrated the new numGrad-CAM method, which generates heatmaps with overlaid numerical focus values for different image regions. Evaluations included technical metrics across 20 runs and subjective assessment by physicians comparing numGrad-CAM to existing CAM methods.	numGrad-CAM-CNN achieved 97.11% accuracy, 95.58% sensitivity, 96.81% specificity for tumor diagnosis. numGrad-CAM explanations were rated highly by 15 physicians for interpretability and usefulness, often preferred over default CAM, Grad-CAM, and Grad-CAM++. Compared to other CAM variants, numGrad-CAM provided close or better accuracy (92.11%).	Validated only on brain tumor MRI with a limited dataset (BRATS 2017, 200 images). Broader applicability to other medical tasks or datasets not tested. Only CNN backbone was explored (no hybrid models or larger architectures).
Ruigang Fu, Qingyong Hu, Xiaohu Dong, Yulan Guo, Yinghui Gao, Biao Li	Axiom-based Grad-CAM (XGrad-CAM) - Proposes an enhanced Grad-CAM method for CNN visualization and explanation by introducing the axioms of Sensitivity and Conservation, aiming for more accurate and theoretically reliable heatmaps.	Introduces two formal axioms (Sensitivity: score change reflects feature importance; Conservation: explanation sums match output) for class activation mapping. Derives XGrad-CAM as a weighted linear combination of feature maps using optimal weights calculated by minimizing violation of these axioms. Validates on various CNN architectures and compares class-discriminability, localization, and efficiency against Grad-CAM, Grad-CAM++, and Ablation-CAM.	XGrad-CAM improves both sensitivity (0.085) and conservation (0.051) over Grad-CAM and Grad-CAM++ and matches or surpasses them in class-discriminability and localization while being much more efficient than Ablation-CAM. Achieves better class-specific heatmaps and covers objects of interest more completely.	Still uses a linear combination of feature maps, not fully addressing non-linear interactions. Gains over Grad-CAM can be dataset/model dependent. Identical results as Grad-CAM and Ablation-CAM on CNNs with global average pooling. Requires additional theoretical setup and optimization over standard Grad-CAM.

XAI findings				
Marco Tulio Ribeiro, Sameer Singh, Carlos Guestrin	LIME (Local Interpretable Model-agnostic Explanations) - Proposes a technique to explain individual predictions of any classifier/regressor by fitting interpretable models locally around each prediction, enhancing trust and understanding in black-box models.	For each prediction, LIME perturbs input data locally and fits a sparse, interpretable linear model weighted by proximity to the original instance. This model highlights features contributing to each prediction. SP-LIME selects representative explanations over a dataset via submodular optimization for model-level insights.	LIME produces faithful, human-interpretable explanations for text and image classifiers (including neural networks). Experiments with simulated and real users show LIME helps in trusting predictions, choosing between models, improving untrustworthy classifiers, and uncovering spurious correlations.	Explanation quality depends on locality of model behavior and choice of interpretable representation. Sparse linear models may not adequately explain highly nonlinear localities. Computational cost increases with sample size and model complexity, especially for images.
Scott M. Lundberg, Su-In Lee	SHAP: A Unified Approach to Interpreting Model Predictions – Proposes SHAP values, which attribute feature importance to each model prediction based on game-theoretic Shapley values, unifying multiple existing explanation methods under a single additive framework.	Defines additive feature attribution methods where explanations are linear sums of feature contributions. Proves Shapley values uniquely satisfy local accuracy, missingness, and consistency axioms. Derives efficient model-agnostic (Kernel SHAP) and model-specific (e.g., Deep SHAP) approximations to compute feature attributions, connecting and improving upon LIME.	SHAP values provide consistent, accurate, and locally faithful feature attributions, matching human intuition better than LIME and DeepLIFT in user studies. Kernel SHAP achieves higher sample efficiency than LIME and Shapley sampling, with improved convergence and computational speed. Demonstrated better performance on text and image models (e.g., MNIST).	Exact computation is exponential in feature number; practical methods rely on approximations. Performance depends on independence and linearity assumptions for feature expectations. Computational cost can be high for large or highly correlated input spaces. Requires careful choice of background/reference data.
Zachary Papanastasopoulos, Ravi K. Samala, Heang-Ping Chan, Lubomir Hadjiiski, Chintana Paramagul, Mark A. Helvie, Colleen H. Neal	Applied explainable AI (XAI) visualization techniques to a dual-domain deep CNN for estrogen receptor (ER) classification from breast DCE-MRI, providing insights into the features learned and identifying both relevant and potentially misleading preprocessing artifacts.	Created an ensemble CNN architecture (based on AlexNet) with spatial and dynamic MRI components, trained using transfer learning on 148 patients' DCE-MRI data. Used Integrated Gradients and SmoothGrad for attribution maps to highlight influential features affecting the model's output.	Model achieved AUC of 0.67 for ER status classification. Attribution maps revealed that the network learned from both tumoral/peritumoral tissue and dynamic signal changes, but also inadvertently from irrelevant preprocessing artifacts (such as image edges). Visualization helped diagnose and suggest improvements in preprocessing steps and model training strategies.	Small dataset may constrain generalizability and performance. Attribution maps sometimes highlighted irrelevant features due to preprocessing errors. Model architecture and findings specific to ER breast MRI.

Shagufta Iftikhar, Nadeem Anjum, Abdul Basit Siddiqui, Masood Ur Rehman, Naeem Ramzan	Proposes an interpretable, simplified CNN model for multi-class brain tumor detection using MRIs, integrating XAI techniques (Grad-CAM, SHAP, LIME) to highlight relevant abnormal tissues and improve transparency in decision-making.	Two large public MRI datasets (Msoud: 7023 images; NeuroMRI: 3264 images) with four classes were preprocessed (cropping, normalization, resizing). A CNN was trained and iteratively refined using Grad-CAM to remove less important layers. Final model explanations were generated using Grad-CAM (spatial heatmaps), SHAP (feature importance), and LIME (local superpixel analysis).	Final XAI-based CNN achieved 99.21% accuracy and F1-score on the main dataset, and 94.72% on the unseen test set. Grad-CAM, SHAP, and LIME visualizations showed the model accurately focused on clinically relevant tumor features, outperforming previous non-XAI and binary approaches.	Validation restricted to MRI data; may not generalize to other modalities (e. g., CT, PET). Only selected XAI methods (Grad-CAM, SHAP, LIME) were evaluated—other explainers not tested. Performance slightly reduced on unseen data due to domain variability.
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